AHSANULLAH UNIVERSITY OF SCIENCE AND TECHNOLOGY

Department of Computer Science and Engineering

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Assignment Question

1. Find all sets of three employees where: A mentors B, B mentors C, C mentors A.

2. Identify patterns where: Employee A reports to B, B reports to C.

3. Identify all pairs where: A mentors B, B does NOT mentor A.

4. Determine the employee(s) with the highest number of incoming mentor edges. Display employee ID, name and total mentees.

5. Find the employee(s) with the lowest number of colleague connections (consider both in-degree and out-degree).

6. Using Label Propagation Algorithm (LPA), detect communities within the employee collaboration network. Display each employee's: Employee ID, Name, Department, Community ID.

7. Compute the PageRank score of each employee and display the top 5 ranked employees. Compare PageRank results with the highest follower (mentor) count. Explain any differences in short.

Answer to the Questions

1. A mentor triangle exists when:

- Employee A mentors B
- Employee B mentors C
- Employee C mentors A

This forms a closed loop of mentorship.

After analyzing the collaboration network, a mentor triangle was identified in the dataset. This indicates a circular mentorship structure where influence flows in a cycle among three employees.

Such a structure suggests strong interconnected collaboration within that group.

2. A valid pattern occurs when:

- A reports_to B
- B reports_to C

This represents a hierarchical reporting structure inside the organization.

The analysis successfully identified reporting chains in the network, confirming the presence of multi-level management hierarchy.

This demonstrates that the organizational structure contains layered supervision relationships.

3.

- Selected all mentor relationships
- Checked whether the reverse relationship exists
- Removed mutual mentorship pairs

The result shows all **one-way mentorship pairs**, meaning:

- A mentors B
- B does not mentor A

This confirms that most mentorship relationships in the network are directional and not reciprocal.

Such asymmetric relationships are realistic in professional environments where senior employees mentor juniors.

4. we calculated the **in-degree of mentor edges** for each employee.

Incoming mentor edges represent the number of mentees an employee has.

After counting the mentor in-degree:

- The employee(s) with the highest number of incoming mentor edges were identified.
- The output displayed:
    - Employee ID
    - Employee Name
    - Total number of mentees

This employee can be considered the most active mentor in terms of mentorship volume.

However, this metric measures only direct mentorship count and does not consider overall influence in the network.

5. For colleague relationships:

- Both incoming and outgoing colleague edges were counted.
- Total colleague degree = in-degree + out-degree.

After computing the degree for each employee, the employee(s) with the minimum colleague connections were identified.

This may indicate:

- Limited peer collaboration

- Departmental isolation
- Fewer professional interactions

Such analysis helps identify employees who are less integrated within the peer collaboration structure.

6. The Label Propagation Algorithm (LPA) was applied to detect communities in the collaboration graph.

How LPA works:

1. Each vertex initially gets a unique label.
2. During each iteration, a vertex adopts the most frequent label among its neighbors.
3. After several iterations, stable communities emerge.

The output displayed:

- Employee ID
- Name
- Department
- Community ID

The results showed that employees who are more densely connected tend to belong to the same community.

In many cases, communities correspond closely to departmental grouping, showing that employees within the same department collaborate more frequently.

7. PageRank was computed for all employees to measure their global importance in the network.

Unlike simple mentor count, PageRank considers:

- Number of incoming edges
- Importance of the employees providing those edges

The top 5 employees were displayed based on their PageRank scores.

Comparison with Highest Mentor Count

It was observed that:

- The employee with the highest number of mentees does not always have the highest PageRank score.

Reason:

Mentor count measures only the quantity of direct mentees (local influence).
PageRank measures global influence, meaning an employee connected to highly influential employees may receive a higher rank even with fewer direct mentees.

Therefore:

- Mentor Count = Local popularity
- PageRank = Global importance

This explains the difference between the two results.

Conclusion:

Through this analysis, the employee collaboration network was successfully modeled and examined using graph analytics techniques.

Pattern detection, degree analysis, community detection, and PageRank provided valuable insights into mentorship structure, hierarchy, collaboration strength, and influence distribution within the organization.

This demonstrates the effectiveness of Apache Spark and GraphX in analyzing complex organizational networks.